

Topology-GRPO (Revised Version)

Abstract

This paper revises the Topology-GRPO framework by introducing an adaptive message fusion mechanism controlled by macro effective duality coupling. The design of inter-group non-commutativity is philosophically inspired by Self-Entangled Duality Dynamics (SED), but no rigorous mathematical derivation connects SED’s quantum spacetime algebra to the present macroscopic engineering framework. All technical components are independent assumptions validated by task performance.

1 Revised Topology-GRPO Framework

1.1 Three-Component Topological Advantage

The three-component topological advantage is calculated purely via reward normalization without intra-group critics, strictly following the core principle of Group Relative Policy Optimization (GRPO). For each group g , the topological advantage is defined as:

$$\begin{aligned}\hat{A}_i^g = & \alpha \cdot \frac{R_i^{\text{global}} - \bar{R}^{\text{global}}}{\sigma^{\text{global}}} \\ & + (1 - \alpha) \cdot \frac{r_i^g - \bar{r}^g}{\sigma^g} \\ & + \beta \cdot \frac{r_i^g - \bar{r}^{\mathcal{N}(g)}}{\sigma^{\mathcal{N}(g)}}.\end{aligned}\tag{1}$$

1.2 Macro Effective Duality Coupling and Adaptive Message Fusion

Motivation Inter-group information propagation in graph-structured systems exhibits two distinct operating regimes: weak coupling with classical commutative fusion, and strong coherent coupling where non-commutative interaction emerges. We introduce the macro effective duality coupling $\mathcal{C}_{\text{eff}}$ to dynamically distinguish these two regimes and switch fusion strategies accordingly.

Remark The proposed $\mathcal{C}_{\text{eff}}$ shares conceptual inspiration with the duality coupling in Self-Entangled Duality Dynamics (SED). However, $\mathcal{C}_{\text{eff}}$ is an independent macroscopic engineering parameter. There is no strict mathematical derivation between SED’s quantum spacetime algebra and the present framework. Their mathematical domains, physical dimensions and research objects are fundamentally different.

Definition of $\mathcal{C}_{\text{eff}}^{(g)}$ Let $\tau_{\text{coh}}^{(g)}$ denote the local policy coherent steps of group g , $\tau_{\text{delay}}^{(g)}$ denote the average inter-group message delay steps, W_{gh} denote the topological coupling weight between group g and neighbor group h , and $\text{Var}(r_h)$ denote the reward variance of neighbor h . $\sigma^{\mathcal{N}(g)}$ is the reward standard deviation of the neighbor set from Eq. (1), and $\epsilon = 10^{-8}$ is a small numerical stabilizer to avoid division by zero. The normalized macro effective duality coupling is formulated as:

$$\mathcal{C}_{\text{eff}}^{(g)} = \frac{\tau_{\text{coh}}^{(g)}}{\tau_{\text{delay}}^{(g)}} \cdot \frac{1}{|\mathcal{N}(g)|} \sum_{h \in \mathcal{N}(g)} W_{gh} \cdot \frac{\text{Var}(r_h)}{(\sigma^{\mathcal{N}(g)})^2 + \epsilon}.\tag{2}$$

Critical Threshold and Operating Regimes $\mathcal{C}_c$ is the critical coupling threshold, which can be set as a fixed hyperparameter or optimized as a learnable parameter during training. Two operating regimes are divided by $\mathcal{C}_c$:

- $\mathcal{C}_{\text{eff}}^{(g)} < \mathcal{C}_c$: **Weak coupling regime**. Adopt classical CONCAT+MLP message fusion.
- $\mathcal{C}_{\text{eff}}^{(g)} \geq \mathcal{C}_c$: **Strong coherent regime**. Activate the q-deformed Q-Fuse fusion module.

Adaptive Fusion Operator Let s_g be the state of group g and m_g be the aggregated inter-group message. The adaptive fusion rule is given by:

$$\tilde{s}_g = \begin{cases} \text{CONCAT}(s_g, \sigma(\text{MLP}[s_g; m_g]) \odot m_g) & \mathcal{C}_{\text{eff}}^{(g)} < \mathcal{C}_c \\ \text{Q-Fuse}(s_g, m_g; \tilde{\theta}) & \mathcal{C}_{\text{eff}}^{(g)} \geq \mathcal{C}_c \end{cases} \quad (3)$$

Q-Fuse: q-Deformed Non-Commutative Fusion Let $\phi(\cdot) = W_\phi \cdot + b_\phi$ and $\psi(\cdot) = W_\psi \cdot + b_\psi$ be two learnable linear transformations. The q-deformed fusion operator is defined as:

$$\text{Q-Fuse}(s_g, m_g; \tilde{\theta}) = W_1 s_g + W_2 m_g + \cos(2\pi\tilde{\theta}) \cdot (\phi(s_g) \odot \psi(m_g) - \phi(m_g) \odot \psi(s_g)). \quad (4)$$

The non-commutative property can be verified directly:

$$\text{Q-Fuse}(m_g, s_g) - \text{Q-Fuse}(s_g, m_g) = 2 \cos(2\pi\tilde{\theta}) (\phi(m_g) \odot \psi(s_g) - \phi(s_g) \odot \psi(m_g)) \neq 0. \quad (5)$$

When $\tilde{\theta} \rightarrow 0$ or the system falls into the weak coupling regime, the non-commutative term vanishes, and Q-Fuse degenerates into classical linear fusion.

Adaptive Update for $\tilde{\theta}$ The twist angle $\tilde{\theta}$ is optimized independently of policy networks, and its update is only activated in the strong coherent regime:

$$\tilde{\theta}(t+1) = \tilde{\theta}(t) + \eta_\theta \cdot \mathbb{1}_{\{\mathcal{C}_{\text{eff}}^{(g)} \geq \mathcal{C}_c\}} \cdot (\mathcal{C}_{\text{eff}}^{(g)} - \mathcal{C}_c) \cdot \frac{\partial \|\text{Q-Fuse}(s_g, m_g) - \text{Q-Fuse}(m_g, s_g)\|^2}{\partial \tilde{\theta}}. \quad (6)$$

where η_θ is the learning rate for $\tilde{\theta}$. The update magnitude is proportional to the distance from the critical threshold $\mathcal{C}_c$.

1.3 Overall Objective Function

The total loss function inherits the original Topology-GRPO formulation, including clipped policy loss, inter-group consistency loss and KL divergence regularization:

$$\begin{aligned} \mathcal{L}^{\text{Topo-GRPO}} = & -\mathbb{E} \left[\min \left(r_t(\theta) \hat{A}^g, \text{clip}(r_t, 1 \pm \epsilon) \hat{A}^g \right) \right] \\ & + \lambda_{\text{cons}} \mathcal{L}_{\text{cons}} + \beta D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}}). \end{aligned} \quad (7)$$

2 Training Algorithm

The complete training pipeline for the updated Topology-GRPO is summarized in Algorithm 1.

3 Compatibility and Ablation Study

3.1 Backward Compatibility

The proposed adaptive module is fully backward compatible with the original Topology-GRPO. When $\mathcal{C}_{\text{eff}}^{(g)} \rightarrow 0$ or $\tilde{\theta} \rightarrow 0$, the non-commutative term is suppressed, and the entire framework degenerates to the original version. All existing experimental settings can be reused directly.

Algorithm 1 Updated Revised Topology-GRPO Training

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1: Initialize high-level policy  $\pi_h$ , group policies  $\{\pi_l^{(g)}\}$  and message encoders
2: Initialize global value network  $V_{\text{global}}$  for topological calibration
3: Initialize hyperparameters:  $\alpha, \beta, \epsilon, \lambda_{\text{cons}}$ 
4: Initialize critical threshold  $\mathcal{C}_c$  (fixed or learnable), initial twist angle  $\tilde{\theta}^{(0)}$  and learning rate  $\eta_\theta$ 
5: while training not converged do
6:   Construct intra-group edges  $E_{\text{intra}}$  and inter-group edges  $E_{\text{inter}}$ 
7:   Collect parallel trajectories for all groups
8:   for each group  $g$  do
9:     Compute  $\tau_{\text{coh}}^{(g)}, \tau_{\text{delay}}^{(g)}$ , neighbor reward  $\text{Var}(r_h)$  and  $\sigma^{\mathcal{N}(g)}$ 
10:    Calculate  $\mathcal{C}_{\text{eff}}^{(g)}$  via Eq. (2)
11:    Obtain aggregated inter-group message  $m_g$  via message passing
12:    Compute fused state  $\tilde{s}_g$  via Eq. (3)
13:    Calculate topological advantage  $\hat{A}^g$  via Eq. (1)
14:  end for
15:  Compute policy loss, inter-group consistency loss  $\mathcal{L}_{\text{cons}}$  and KL divergence
16:  Update  $\pi_h$  and  $\{\pi_l^{(g)}\}$  by minimizing  $\mathcal{L}^{\text{Topo-GRPO}}$ 
17:  if  $\exists g, \mathcal{C}_{\text{eff}}^{(g)} \geq \mathcal{C}_c$  then
18:    Update  $\tilde{\theta}$  via Eq. (6)
19:  end if
20:  Gradually decay  $\lambda_{\text{cons}}$ 
21: end while
22: return  $\pi_h, \{\pi_l^{(g)}\}$ 
```

3.2 Ablation Design

To quantitatively evaluate the contribution of each component, we design the following ablation groups:

1. Baseline: Original Topology-GRPO ($\mathcal{C}_{\text{eff}}$ and Q-Fuse removed)
2. $\mathcal{C}_{\text{eff}}$ only: Adaptive threshold enabled, always use classical CONCAT+MLP fusion
3. Full model: $\mathcal{C}_{\text{eff}}$ + Q-Fuse + decoupled $\tilde{\theta}$ update
4. Ablate non-commutativity: Set $\cos(2\pi\tilde{\theta}) \equiv 0$ to disable asymmetric terms
5. Fixed $\mathcal{C}_c$ vs. learnable $\mathcal{C}_c$: Compare two threshold settings